

Project Abstract

This BI architecture, as sketched in Figure 1, is mostly related to the mixed data warehouse house that involves the core data warehouse, data marts, and the virtual data warehouse. In addition to the core data warehouse, the KNIME data mining tools provide activities related to data mart, virtual data warehouses, and data mining. Aside from the data generated from the management of the customer relationship, enterprise resource planning, and supply chain management. These are termed internal data. Reviews and other social media data also come in as external data. HevoData is used to extract, transform, and load (ETL) the data from these internal and external sources into the core data warehouse of the bank. This tool extracted the sourced data to the staging area, cleaned it, aggregated it, and harmonized it into a relational model, then loaded it into the database. The chosen data mining tool, KNIME, can also perform ETL, but for efficiency, I prefer to utilize it after the core data warehouse for multi-dimensional modeling, predictive modeling, other business data analytics, and reporting through its various nodes and plugins for other tool integration.

KNIME integration nodes enhance the integration of the Weka tools to explore its wide variety of machine learning algorithms for predictive data mining. Additionally, to explore virtual data warehouses, the KNIME server is integrated with the cloud through the REST API; this will enhance the real-time analytics ability of the tool. Other tasks carried out within the KNIME data mining too are

1. The KNIME server reads and rewrites the data received from the core data warehouse into multidimensional models from the data
2. Enable the creation of workflows for every predictive modelling and other business intelligence solutions to business growth, especially customer acquisition and retention.
3. Ensure the deployment of the predictive models through deployment nodes for periodic modelling of the predictive analytics to maintain business advantage.
4. Prepare a real-time accurate and precise report of the business analytics to guide the management in the choice of the business decision.

Summarily, sequel to the three questions concerned with this project portfolio, the continuous mining of the data will provide visualized charts and reports that will guide the management of proactive steps to take in every case.

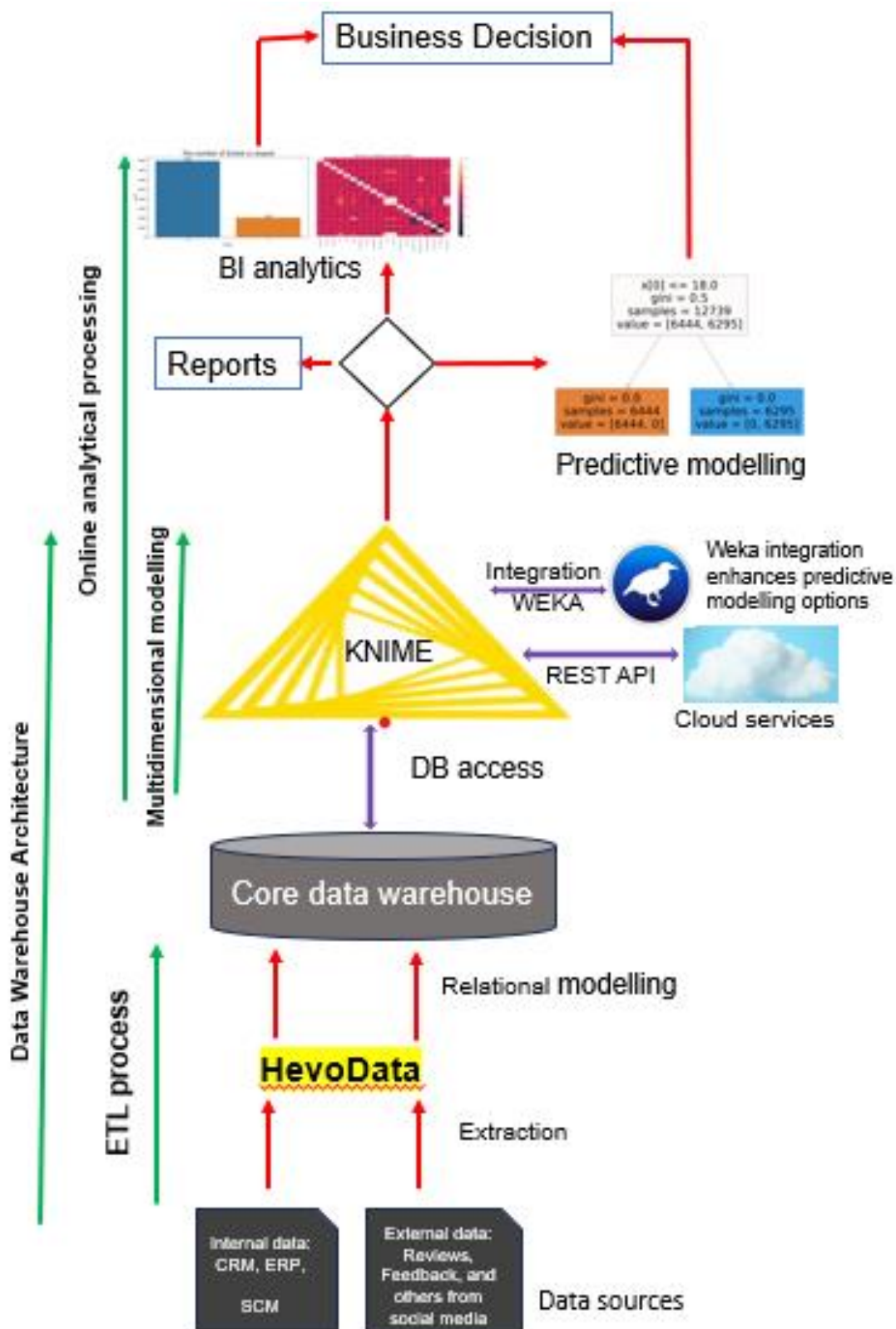


Figure 1 shows the business intelligence architecture of the company with the integrated data mining tool. (Own Illustration)